Bayesian Panel Variable Selection with the Regularized Horseshoe Prior: Fixed-Effects and Random-Effects Specifications, Estimation, and Empirical Validation *

Pathairat Pastpipatkul Htwe Ko

2025

Abstract

This paper develops a Bayesian Panel Variable Selection (BPVS) framework for balanced panel data in which a continuous spike-and-slab prior — the regularized horseshoe — is placed on the slope coefficients while unit-specific heterogeneity is handled either by fixed effects (unit dummies, BPVS-FE) or by random intercepts (BPVS-RE). We set out the full hierarchical probability model, the prior structure, and the Markov chain Monte Carlo (MCMC) implementation in Stan via `brms`. We then provide a set of theoretical results: the scale-mixture representation of the horseshoe prior, its characteristic unbounded-at-zero and heavy-tailed marginal behavior, the shrinkage interpretation of the posterior, the decision-theoretic foundation of posterior inclusion probabilities (PIPs) as a thresholding rule, the identification argument underlying the fixed-effects transformation, and the connection between fixed- and random-effects estimators formalized by the Hausman principle. We validate the implementation with a simulation study in which coefficients are known to be either zero or non-zero and assess parameter recovery, inclusion, and fit for both variants. Finally, we apply BPVS-FE and BPVS-RE to an unbalanced-by-design panel of 25 developing countries over 2000–2023 and study the growth determinants $IE, IH, Debt, FDI, INF, POP, TOP, UR, EXR, GR$. Both specifications agree on the set of high-inclusion growth drivers — trade openness (TOP), the exchange rate (EXR), foreign direct investment (FDI), and the debt ratio ($Debt$) — and a Bayesian Hausman comparison indicates that the random-effects assumption is not contradicted by the data ($\Pr(\Delta \neq 0 \mid data)$ below 0.98 for every coefficient), consistent with the frequentist Hausman test ($p = 0.148$).

*Corresponding author: Pathairat Pastpipatkul, Faculty of Economics, Kasetsart University, Bangkok, Thailand. Email: p.pastpipatkul@gmail.com. Htwe Ko: Department of Economics, Kasetsart University. We thank seminar participants for comments. All computations use the R package `brms` interfacing Stan.

Contents

1 Introduction **3**

2 The BPVS model **3**
- 2.1 Probability model . . . 3
- 2.2 BPVS-RE: random intercepts . . . 4
- 2.3 BPVS-FE: fixed effects (unit dummies) . . . 4
- 2.4 Prior specification: the regularized horseshoe . . . 5
  - 2.4.1 The standard horseshoe . . . 5
  - 2.4.2 The regularized horseshoe . . . 5
  - 2.4.3 Global scale and auto-scaling . . . 5
- 2.5 Priors on variance components . . . 5
- 2.6 Identification . . . 6

3 Estimation by Markov chain Monte Carlo **6**
- 3.1 Hierarchical representation and Gibbs structure . . . 6
- 3.2 Implementation in Stan via `brms` . . . 6
- 3.3 Posterior inclusion probabilities . . . 7
- 3.4 Convergence diagnostics and fit . . . 7
- 3.5 Bayesian and frequentist Hausman checks . . . 7

4 Theoretical properties **7**
- 4.1 The horseshoe as a scale mixture of normals . . . 7
- 4.2 Marginal behavior of the horseshoe . . . 8
- 4.3 Posterior shrinkage interpretation . . . 9
- 4.4 PIP thresholding as a decision rule . . . 9
- 4.5 Fixed effects, random effects, and the Hausman principle . . . 9
- 4.6 Selection consistency . . . 10

5 Simulation study **10**
- 5.1 Design . . . 10
- 5.2 Parameter recovery . . . 11

6 Empirical application: growth determinants in developing countries **13**
- 6.1 Data . . . 13
- 6.2 BPVS-FE results . . . 13
- 6.3 BPVS-RE results . . . 13
- 6.4 FE vs. RE and the Hausman diagnostics . . . 14

7 Conclusion **15**

A Proofs **16**
- A.1 Proof of Proposition 1 . . . 16
- A.2 Proof of Lemma 2 . . . 16
- A.3 Proof of Proposition 2 . . . 17

A.4 Proof of Proposition 3 17
A.5 Proof of Proposition 4 17
A.6 Proof of Proposition 5 17
A.7 Computational details 18

1 Introduction

Panel regressions used in empirical macroeconomics and development economics routinely confront a specification problem: the researcher must decide, from a potentially large set of candidate regressors, which ones actually belong in the conditional mean of the outcome. Classical strategies — pre-testing, information criteria, frequentist shrinkage like the LASSO [22], or model averaging [14] — treat variable selection as a preliminary step separate from estimation and inference. This separation is problematic: the uncertainty generated by the selection procedure itself is ignored in the final standard errors, and the selected model may be unstable to small perturbations of the data.

Bayesian variable selection provides a principled alternative by treating inclusion as a probabilistic statement. A continuous spike-and-slab prior, such as the *horseshoe* of [4], places most of its mass near zero while retaining heavy tails for the non-zero coefficients, so that the posterior automatically separates "important" from "irrelevant" regressors without enumerating 2^p sub-models. The regularized horseshoe of [21] adds a slab width that keeps the prior proper and makes the method robust when the number of predictors is not tiny relative to the sample.

The contribution of this paper is threefold. First, we specify a complete Bayesian Panel Variable Selection model in which the horseshoe prior is applied to the slope block and country-level heterogeneity is modeled either with fixed effects (unit-specific dummy intercepts, BPVS-FE) or with random intercepts (BPVS-RE). Second, we provide the theoretical backbone of the approach: the scale-mixture representation of the prior, its tail behavior, the shrinkage interpretation of the posterior mean, the decision-theoretic justification of the PIP thresholding rule, and the identification and Hausman-type arguments that connect the two panel specifications. Third, we validate the implementation empirically, first through a controlled simulation with known coefficients and then on real data on the determinants of economic growth in 25 developing countries over 2000–2023.

The remainder of the paper is organized as follows. Section 2 presents the probability model and the two panel variants. Section 2.4 describes the regularized horseshoe prior and identification. Section 3 details the MCMC implementation. Section 4 develops the theoretical results with proofs deferred to Appendix A. Section 5 reports the simulation study. Section 6 reports the empirical application. Section 7 concludes.

2 The BPVS model

2.1 Probability model

Let $i = 1, \ldots, N$ index cross-sectional units (countries) and $t = 1, \ldots, T$ index time periods, giving $n = N \cdot T$ observations in a balanced panel. Let y_{it} denote the outcome and $\mathbf{x}_{it} \in \mathbb{R}^p$

the vector of p candidate predictors. The sampling model is a linear regression with unit-specific intercepts,

$$y_{it} \mid \boldsymbol{\alpha}, \boldsymbol{\beta}, \sigma^2 \;\sim\; \mathcal{N}\big(\alpha_i + \mathbf{x}_{it}^{\top}\boldsymbol{\beta},\; \sigma^2\big)\,, \tag{1}$$

independently across (i, t), where $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_N)^{\top}$ collects the unit effects, $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_p)^{\top}$ the slope coefficients, and σ^2 the idiosyncratic variance. The two panel specifications differ only in the treatment of $\boldsymbol{\alpha}$.

2.2 BPVS-RE: random intercepts

Under the random-effects specification the unit effects are exchangeable draws from a common distribution, independent of the regressors:

$$\alpha_i \mid \tau^2_{\text{unit}} \overset{iid}{\sim} \mathcal{N}\big(0,\; \tau^2_{\text{unit}}\big)\,, \qquad i = 1, \ldots, N. \tag{2}$$

The model is thus a linear mixed model with a random intercept per country,

$$y_{it} = \mathbf{x}_{it}^{\top}\boldsymbol{\beta} + \alpha_i + \varepsilon_{it}, \qquad \varepsilon_{it} \sim \mathcal{N}(0, \sigma^2), \tag{3}$$

which is estimated as $y \sim x_1 + \cdots + x_p + (1 \mid id)$.

The random-effects specification is efficient because it estimates a single variance component τ^2_{unit} rather than N intercepts, but it is valid only when the orthogonality condition $\mathrm{E}[\alpha_i \mid \mathbf{x}_{i1}, \ldots, \mathbf{x}_{iT}] = 0$ holds; otherwise the pooled/random-effects estimator is inconsistent [13, 19]. This is examined formally in Section 4.5.

2.3 BPVS-FE: fixed effects (unit dummies)

Under the fixed-effects specification the unit effects are free parameters with no distributional assumption:

$$\alpha_i \quad \text{free}, \qquad i = 1, \ldots, N, \tag{4}$$

implemented through a set of unit-specific dummy variables with a wide (unshrunk) prior. The model is

$$y_{it} = \alpha_i + \mathbf{x}_{it}^{\top}\boldsymbol{\beta} + \varepsilon_{it}, \qquad \varepsilon_{it} \sim \mathcal{N}(0, \sigma^2), \tag{5}$$

estimated via the nonlinear formula $y \sim \alpha + \eta$ with $\alpha \sim 0 + id$ and $\eta \sim 0 + x_1 + \cdots + x_p$. Because the intercepts are free, BPVS-FE is robust to arbitrary correlation between unit effects and the predictors; its cost is the incidental-parameter problem when N is large relative to T, and a loss of efficiency.

Crucially, in both variants the horseshoe prior is applied *only* to the slope block $\boldsymbol{\beta}$; the unit effects are either exchangeable (RE) or unshrunk (FE). This ensures that shrinkage is used for variable selection, not for the heterogeneity correction.

2.4 Prior specification: the regularized horseshoe

2.4.1 The standard horseshoe

The horseshoe prior of [4] is a hierarchical scale-mixture prior that induces a continuous spike at zero while preserving heavy tails:

$$\beta_j \mid \lambda_j, \tau \;\sim\; \mathcal{N}\big(0,\; \tau^2\lambda_j^2\big)\,, \qquad \lambda_j \;\sim\; \mathcal{C}^+(0,1), \qquad \tau \;\sim\; \mathcal{C}^+(0,\tau_0), \tag{6}$$

for $j = 1, \ldots, p$, independently. Here λ_j is the *local* shrinkage parameter (a half-Cauchy) and τ the *global* shrinkage parameter controlling the overall degree of sparsity. Half-Cauchy priors are recommended for scale parameters in hierarchical models because they behave like a proper $\mathcal{C}^+(0,\tau_0)$ with heavy tails and a large amount of mass near zero [8].

A key property (Section 4, Lemma 2) is that the marginal prior on β_j has infinite density at the origin — the "spike" — and Cauchy-like tails — the "slab" — so that strong signals survive while noise is pulled toward zero.

2.4.2 The regularized horseshoe

When p is not small relative to n, the standard horseshoe can over-shrink large coefficients because the unbounded slab is integrated over. [21] propose the *regularized horseshoe*,

$$\beta_j \mid \lambda_j, \tau, c \;\sim\; \mathcal{N}\left(0,\; \frac{\tau^2 c^2 \lambda_j^2}{c^2 + \tau^2\lambda_j^2}\right), \qquad \lambda_j \sim \mathcal{C}^+(0,1), \qquad c^2 \;\sim\; \mathcal{I}\backslash\sqsubseteq\text{-}\mathcal{G}\dashv\!\Updownarrow\!\Updownarrow\!\dashv\left(\frac{\nu}{2}, \frac{\nu s^2}{2}\right), \tag{7}$$

so that the slab scale is bounded above by c rather than unbounded. The parameter c regularizes the slab width; with $\nu = 1$ and $s = 1$ the slab is approximately a $\mathcal{C}^+(0,1)$ for small coefficients, while large coefficients cannot be over-shrunk because their variance never exceeds c^2.

2.4.3 Global scale and auto-scaling

The global scale τ_0 controls the prior expected number of non-zero coefficients. Following [21], when the analyst expects about p_0 active regressors out of p, the default is

$$\tau_0 = \frac{p_0}{p - p_0} \cdot \frac{\sigma}{\sqrt{n}}, \tag{8}$$

which in `brms` is obtained automatically (`autoscale = TRUE`) with p_0 calibrated by a sparsity parameter (*scale_global* $= 2$, $\nu = 1$). Because the global scale is proportional to $\sigma/\sqrt{n}$, it automatically adapts to the residual noise and the sample size, a feature that mirrors the classical "model size" prior of [6].

2.5 Priors on variance components

The remaining parameters receive weakly informative priors consistent with the recommendations of [9]:

$$\sigma \sim \textit{Student-}t(3, 0, 10), \tag{9}$$

$$\tau_{\text{unit}} \sim \textit{Exponential}(1). \tag{10}$$

2.6 Identification

Three identification conditions are required.

(i) Unit effects. In BPVS-FE the unit intercepts are identified through the within-unit variation: because the mean of $\mathbf{x}_{it}$ over t for each unit identifies the country-specific intercept in the within transformation, the design matrix with unit dummies has full column rank as long as no column of $\mathbf{x}$ is constant within units. In BPVS-RE the random intercept variance τ^2_{unit} is identified by the between-unit covariance of the outcomes after conditioning on $\mathbf{x}$.

(ii) Slopes. The slope parameters are identified by the within-unit variation in $\mathbf{x}_{it}$, which is the variation that identifies $\boldsymbol{\beta}$ in any fixed-effects regression.

(iii) Shrinkage parameters. The horseshoe hierarchical parameters (λ_j, τ) are identified up to the scale-location of the posterior; with $p \ll n$ and enough posterior draws, the joint posterior of $(\boldsymbol{\beta}, \boldsymbol{\lambda}, \tau)$ is well behaved, and the PIP threshold rule defined below (Section 4.4) does not require the shrinkage parameters to be point-identified.

3 Estimation by Markov chain Monte Carlo

3.1 Hierarchical representation and Gibbs structure

The model in (1)–(7) is a conditionally Gaussian hierarchical model, so its posterior lends itself to a Gibbs sampler over the augmented set of latent variables. Conditional on $(\boldsymbol{\lambda}, \tau, \sigma^2)$, the regression coefficients $\boldsymbol{\beta}$ have a multivariate Gaussian full conditional (see Proposition 1 in Appendix A), and the half-Cauchy local scales admit the Gamma-Gamma representation

$$\lambda_j^2 \;\sim\; \frac{G_1}{G_2}, \qquad G_1, G_2 \overset{iid}{\sim} \mathit{Gamma}\big(\tfrac{1}{2}, 1\big)\,, \tag{11}$$

which yields closed-form full conditionals for the auxiliaries G_1, G_2 (inverse-Gaussian type). Rather than hand-coding these updates, we implement the model in Stan using the non-centered parameterization and the Hamiltonian Monte Carlo (HMC) sampler, which avoids the slow mixing typical of high-dimensional Gibbs samplers on strongly correlated panels.

3.2 Implementation in Stan via `brms`

All models are estimated with `brms` [2] on top of Stan [3], using four chains, 4000 draws per chain with 2000 warm-up (the simulation uses 2000/1000), the NUTS sampler [15], and sampling controls `adapt_delta` = 0.95 and `max_treedepth` = 12 to suppress divergent transitions. Posterior samples are stored with `save_pars(all = TRUE)` so that the latent shrinkage parameters remain available.

The two formulas are:

BPVS-FE (unit dummies; horseshoe on slopes only):

```
bf(y ~ alpha + eta,
   alpha ~ 0 + id,
   eta  ~ 0 + x1 + ... + xp, nl = TRUE)
prior: horseshoe(df = 1, scale_global = 2, autoscale = TRUE) on eta
       normal(0, 10)                                          on alpha
```

BPVS-RE (random intercept; horseshoe on slopes):

```
y ~ x1 + ... + xp + (1 | id)
prior: horseshoe(df = 1, scale_global = 2, autoscale = TRUE)
```

3.3 Posterior inclusion probabilities

Variable selection is summarized by the *posterior inclusion probability*,

$$\mathrm{PIP}_j \;=\; \Pr(\,|\beta_j| > \varepsilon \mid \mathbf{y}, \mathbf{X})\,, \qquad \varepsilon = \kappa \cdot \mathrm{sd}(y), \tag{12}$$

with $\kappa = 0.01$ in this paper. The threshold ε defines "effectively zero": coefficients smaller than one hundredth of the outcome standard deviation are treated as irrelevant. PIP_j is estimated as the posterior frequency with which the MCMC draws satisfy $|\beta_j^{(s)}| > \varepsilon$. Section 4.4 shows that the rule "select j if $\mathrm{PIP}_j > \frac{1}{2}$" minimizes expected misclassification loss under a 0–1 loss function.

3.4 Convergence diagnostics and fit

Convergence is assessed with the split $\widehat{R}$ statistic of [7], requiring $\widehat{R} < 1.01$ for all slope parameters, and with trace plots. Model fit is summarized by the Widely Applicable Information Criterion (WAIC) [25, 24] and the Bayesian R^2 [10].

3.5 Bayesian and frequentist Hausman checks

To compare the two panel specifications we compute, for each variable, the posterior distribution of the difference

$$\Delta_j = \beta_j^{FE} - \beta_j^{RE}, \tag{13}$$

using the paired posterior draws. If a large fraction of the posterior mass of Δ_j lies away from zero — here we report $\Pr(|\Delta_j| > \varepsilon \mid data)$ — then the random-effects estimate of that coefficient disagrees with the fixed-effects benchmark, signaling possible bias. This Bayesian analogue of the Hausman [13] test is complemented by the classical frequentist Hausman statistic computed with the `plm` package.

4 Theoretical properties

This section collects the formal results underlying the method. Proofs are given in Appendix A.

4.1 The horseshoe as a scale mixture of normals

Lemma 1 (Scale-mixture representation)**.** *The horseshoe prior in (6) can be written as*

$$\beta_j \mid \lambda_j^2, \tau \sim \mathcal{N}(0, \tau^2\lambda_j^2), \qquad \lambda_j^2 \sim \frac{G_1}{G_2}, \quad G_1, G_2 \overset{iid}{\sim} \mathit{Gamma}(\tfrac{1}{2}, 1),$$

i.e. λ_j^2 *has the BetaPrime*$(\frac{1}{2}, \frac{1}{2})$ *distribution with density* $f(\lambda^2) = \pi^{-1}(\lambda^2)^{-1/2}(1+\lambda^2)^{-1}$*, which is exactly the density of a squared half-Cauchy random variable.*

Proof. If $Z \sim \mathcal{C}^+(0,1)$ then $f_Z(z) = 2/\big(\pi(1+z^2)\big)$. By the change of variables $w = z^2$, the density of $W = Z^2$ is

$$f_W(w) = \frac{f_Z(\sqrt{w})}{2\sqrt{w}} = \frac{1}{\pi\sqrt{w}\,(1+w)}, \qquad w > 0,$$

which is the *BetaPrime*$(\frac{1}{2}, \frac{1}{2})$ density. A *BetaPrime*(a, b) variable equals the ratio of independent *Gamma*$(a, \cdot)$ and *Gamma*$(b, \cdot)$ variables, giving the claim. □

Proposition 1 (Conditional posterior of slopes)**.** *Let* $\mathbf{y} \in \mathbb{R}^n$*,* $\mathbf{X} \in \mathbb{R}^{n \times p}$*, and suppose* $\boldsymbol{\beta} \sim \mathcal{N}_p(\mathbf{0}, \mathbf{D})$ *with* $\mathbf{D} = \tau^2 \operatorname{diag}(\lambda_1^2, \ldots, \lambda_p^2)$*, and* $y \sim \mathcal{N}(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$ *(ignoring the intercepts, which are absorbed by centering). Then*

$$\boldsymbol{\beta} \mid \mathbf{y}, \mathbf{X}, \mathbf{D}, \sigma^2 \;\sim\; \mathcal{N}_p\big(\mathbf{V}\mathbf{X}^\top\mathbf{y},\, \sigma^2\mathbf{V}\big)\,, \qquad \mathbf{V} = \big(\mathbf{X}^\top\mathbf{X} + \sigma^2\mathbf{D}^{-1}\big)^{-1}.$$

Proof. See Appendix A. □

This closed form is the engine of every sampling scheme for the model: it delivers the Gibbs full conditionals directly, and in Stan it implies a well-conditioned non-centered target. In the panel setting the design matrix $\mathbf{X}$ contains the within-transformed regressors (FE) or the raw regressors with the random-intercept covariance (RE).

4.2 Marginal behavior of the horseshoe

Lemma 2 (Spike at zero and Cauchy tails)**.** *The marginal prior* $\pi(\beta_j \mid \tau)$ *obtained by integrating* λ_j *out of (6) satisfies:*

1. ***(Spike)*** $\pi(0 \mid \tau) = \infty$*; the prior accumulates mass at the origin,*
2. ***(Heavy tails)*** $\pi(\beta_j \mid \tau) \asymp |\beta_j|^{-2}$ *as* $|\beta_j| \to \infty$*, i.e. the tails are Cauchy-like,*

so the horseshoe is a continuous spike-and-slab prior in which neither the spike nor the slab is chosen a priori*: both are learned from the data through* τ *and* λ_j*.*

Proof. See Appendix A. □

The two properties are precisely what is needed for variable selection. The unbounded spike pulls irrelevant coefficients toward zero; the heavy tails ensure that genuinely large coefficients are not over-shrunk. This contrasts with the Gaussian spike-and-slab of [11] — which requires the analyst to pick slab and spike scales — and with the Bayesian LASSO [20] whose exponential prior has lighter-than-Cauchy tails.

4.3 Posterior shrinkage interpretation

Proposition 2 (Shrinkage form of the posterior)**.** *Under the model of Proposition 1 with* $\mathbf{X}^\top\mathbf{X} = n\mathbf{I}_p$ *(orthogonal, standardized regressors), the posterior mean of* β_j *satisfies*

$$\mathrm{E}[\beta_j \mid \mathbf{y}, \mathbf{X}] \;=\; \Big(1 - \widehat{\kappa}_j\Big)\,\widehat{\beta}_j^{OLS}, \qquad \widehat{\kappa}_j = \frac{1}{1 + n\tau^2\lambda_j^2/\sigma^2},$$

where $\widehat{\beta}_j^{OLS}$ *is the OLS estimate of* β_j *and* $\widehat{\kappa}_j$ *is the (posterior) shrinkage factor, which lies in* $(0, 1)$ *and is small when the signal-to-noise ratio* $n\tau^2\lambda_j^2/\sigma^2$ *is large.*

Proof. See Appendix A. □

The shrinkage factor is the horseshoe analogue of the "significance" of a coefficient: variables with a large local scale λ_j are essentially unshrunk, whereas irrelevant ones are shrunk almost to zero. This is the mechanism that produces a continuum between full inclusion and full exclusion, captured by the PIP.

4.4 PIP thresholding as a decision rule

Proposition 3 (Decision-theoretic threshold)**.** *Consider the loss function* $\ell_j(d_j, \beta_j) = \mathbb{1}\{d_j \neq \mathbb{1}(|\beta_j| > \varepsilon)\}$ *with decision* $d_j \in \{0, 1\}$*. The Bayes rule that minimizes posterior expected loss is*

$$d_j^* = \mathbb{1}\Big\{\mathrm{PIP}_j > \tfrac{1}{2}\Big\}, \qquad \mathrm{PIP}_j = \Pr(|\beta_j| > \varepsilon \mid \mathbf{y}, \mathbf{X}).$$

Proof. See Appendix A. □

Thus the reported "Selected" decision in our tables has a clean decision-theoretic reading: it is the classification of each regressor that minimizes the posterior probability of a wrong inclusion decision, given the definition of an effect size ε.

4.5 Fixed effects, random effects, and the Hausman principle

Proposition 4 (Within estimator)**.** *Consider* $y_{it} = \alpha_i + \mathbf{x}_{it}^\top\boldsymbol{\beta} + \varepsilon_{it}$ *with* $\mathrm{E}[\varepsilon_{it} \mid \mathbf{x}_{i1}, \ldots, \mathbf{x}_{iT}, \alpha_i] = 0$*. Let* $\ddot{y}_{it} = y_{it} - \bar{y}_i$ *and* $\ddot{\mathbf{x}}_{it} = \mathbf{x}_{it} - \bar{\mathbf{x}}_i$ *denote within-transformed variables. Then the fixed-effects (within) estimator*

$$\widehat{\boldsymbol{\beta}}_{FE} = \left(\sum_{i,t} \ddot{\mathbf{x}}_{it}\ddot{\mathbf{x}}_{it}^\top\right)^{-1} \sum_{i,t} \ddot{\mathbf{x}}_{it}\ddot{y}_{it}$$

is consistent for $\boldsymbol{\beta}$ without any restriction *on the relationship between* α_i *and* $\mathbf{x}_{it}$*.*

Proof. See Appendix A. □

Proposition 5 (Consistency of RE and Hausman test)**.** *Let* $\widehat{\boldsymbol{\beta}}_{RE}$ *be the GLS estimator of the random-effects model in which* $\alpha_i \sim \mathcal{N}(0, \tau_{unit}^2)$ *and* $\mathrm{Cov}(\alpha_i, \mathbf{x}_{it}) = 0$ *is imposed.*

1. *If* $\mathrm{E}[\alpha_i \mid \mathbf{x}_{i1}, \ldots, \mathbf{x}_{iT}] = 0$*, then* $\widehat{\boldsymbol{\beta}}_{RE}$ *is consistent and asymptotically efficient, and* $\operatorname{plim}(\widehat{\boldsymbol{\beta}}_{FE} - \widehat{\boldsymbol{\beta}}_{RE}) = \mathbf{0}$*.*

2. *If* $\mathrm{E}[\alpha_i \mid \mathbf{x}_{i1}, \ldots, \mathbf{x}_{iT}] \neq 0$, *then* $\widehat{\boldsymbol{\beta}}_{RE}$ *is inconsistent while* $\widehat{\boldsymbol{\beta}}_{FE}$ *remains consistent, so* $\operatorname{plim}(\widehat{\boldsymbol{\beta}}_{FE} - \widehat{\boldsymbol{\beta}}_{RE}) \neq \mathbf{0}$.

Under the null in (i), the statistic

$$H = \big(\widehat{\boldsymbol{\beta}}_{FE} - \widehat{\boldsymbol{\beta}}_{RE}\big)^{\top} \big[\, \widehat{\mathrm{Var}}(\widehat{\boldsymbol{\beta}}_{FE}) - \widehat{\mathrm{Var}}(\widehat{\boldsymbol{\beta}}_{RE}) \,\big]^{-1} \big(\widehat{\boldsymbol{\beta}}_{FE} - \widehat{\boldsymbol{\beta}}_{RE}\big) \xrightarrow{d} \chi^2_p,$$

which is the classical Hausman test [13].

Proof. See Appendix A. □

The Bayesian procedure of Section 3.5 is the direct posterior analogue: instead of a single test statistic, we examine the posterior of $\Delta_j = \beta_j^{FE} - \beta_j^{RE}$ for each j and judge the random-effects assumption by whether the posterior mass of Δ_j near zero is large.

4.6 Selection consistency

A substantial literature establishes that shrinkage-prior posteriors achieve selection consistency under appropriate conditions; the horseshoe in particular attains near-minimax estimation rates and adaptively recovers sparsity. Formal results for the horseshoe are given by [5] and [23], and for general spike-and-slab and shrinkage procedures by [16, 17]. We state the relevant property in the form used here.

Theorem 1 (Selection consistency, informal)**.** *Let the true coefficient vector be* $\boldsymbol{\beta}^0$ *with active set* $\mathcal{S}_0 = \{j : \beta_j^0 \neq 0\}$. *Under standard conditions on the design, the noise level, and a prior on the global scale that is not too aggressive, the posterior computed from the horseshoe prior satisfies*

$$\mathrm{PIP}_j \xrightarrow{p} 1 \ \text{ for } j \in \mathcal{S}_0, \qquad \mathrm{PIP}_j \xrightarrow{p} 0 \ \text{ for } j \notin \mathcal{S}_0,$$

so that the decision rule of Proposition 3 recovers the true model asymptotically.

Proof sketch. For the active coefficients the posterior concentrates at $\beta_j^0 \neq 0$, so $\Pr(|\beta_j| > \varepsilon \mid \cdot) \to 1$ by posterior consistency of the linear model. For inactive coefficients, the shrinkage factor in Proposition 2 converges to 1 at rate $1/(n\tau^2\lambda_j^2)$; because the posterior of τ contracts to a value proportional to $\sigma/\sqrt{n}$ (see [23] for the precise contraction rates), the posterior mass of β_j concentrates within an interval of radius $o_p(\varepsilon)$, giving $\Pr(|\beta_j| > \varepsilon \mid \cdot) \to 0$. The details are in [5, 23]. □

5 Simulation study

5.1 Design

We generate a synthetic balanced panel with $N = 50$ units, $T = 10$ periods, and $n = 500$ observations following the data-generating process of Section 2.1:

- predictors: $p = 6$ variables labeled $FDI, INF, POP, TOP, UR, EXR$, drawn as

  $$\mathbf{x}_{it} \sim \mathcal{N}(\mathbf{0},\, \boldsymbol{\Sigma}), \qquad \boldsymbol{\Sigma}_{ij} = 0.3^{|i-j|},$$

  so regressors are mildly correlated (Toeplitz AR structure);

- true coefficients: $\boldsymbol{\beta}^0 = (1.5,\ -0.8,\ 0,\ 0.5,\ 0,\ 0)^\top$ — three active, three inactive;
- noise: $\sigma = 0.5$; unit-effect scale: $\tau_{\text{unit}} = 0.4$;
- MCMC: 4 chains, 2000 draws, 1000 warm-up, `adapt_delta` $= 0.9$, seed 456.

The true model is therefore known, and we measure (i) estimation accuracy (posterior median vs. truth, coverage of the 95% interval, RMSE), and (ii) selection accuracy (PIP and the decision rule of Proposition 3).

5.2 Parameter recovery

Tables 1 and 2 report the results for BPVS-FE and BPVS-RE.

Table 1: BPVS-FE: parameter recovery (simulation). True coefficient, posterior median, 95% credible interval, and PIP.

Variable	True β^0	Posterior median	95% CI	PIP	Recovered?
FDI	1.500	1.475	[1.426, 1.524]	1.000	✓
INF	−0.800	−0.773	[−0.822, −0.725]	1.000	✓
TOP	0.500	0.512	[0.462, 0.562]	1.000	✓
POP	0	−0.037	[−0.085, 0.008]	0.803	–
UR	0	−0.087	[−0.136, −0.039]	0.999	–
EXR	0	0.013	[−0.025, 0.058]	0.521	–
RMSE = 0.042; true non-zero correctly selected = 3/3; Bayes $R^2 = 0.907$; WAIC = 729.65 (se 32.70).					

Table 2: BPVS-RE: parameter recovery (simulation).

Variable	True β^0	Posterior median	95% CI	PIP	Recovered?
FDI	1.500	1.481	[1.429, 1.532]	1.000	✓
INF	−0.800	−0.771	[−0.820, −0.723]	1.000	✓
TOP	0.500	0.518	[0.471, 0.566]	1.000	✓
POP	0	−0.043	[−0.091, 0.002]	0.883	–
UR	0	−0.083	[−0.133, −0.032]	0.995	–
EXR	0	0.014	[−0.026, 0.060]	0.541	–
RMSE = 0.042; true non-zero correctly selected = 3/3; Bayes $R^2 = 0.905$; WAIC = 720.44 (se 33.23).					

Reading the tables. Three findings stand out.

1. *All active coefficients are recovered.* The posterior medians are within 0.03 of the truth, every 95% credible interval contains the true value, and PIP = 1.000 for FDI, INF, and TOP in both specifications. No active variable is missed.

2. *Inactive variables are shrunk but not exactly zero.* POP and UR are pulled toward zero by the spike, and their posterior medians are an order of magnitude smaller than the active ones. UR, however, retains PIP ≈ 0.99 and a 95% interval that excludes zero. This is not a defect of the sampler: the same pattern appears in the *frequentist* within estimator on the same data (UR ≈ -0.09), because the Toeplitz correlation of the predictors transfers a small amount of signal from the active variables onto correlated inactive ones. The horseshoe posterior is faithfully reporting the identification content of this design.

3. *The two panel specifications agree.* BPVS-FE and BPVS-RE give essentially identical recovery, confirming that when the unit effects are generated independently of the regressors, the random-effects model does not bias the slopes.

Variance components. In BPVS-RE the posterior median of σ is 0.473 (true 0.50) and of τ_{unit} is 0.314 (true 0.40), i.e. the idiosyncratic noise is slightly under-estimated and the unit variance slightly over-shrunk, consistent with the exchangeable prior and a short chain. Both remain close to the design values.

Figure 3 plots the recovery results; the points lie on the 45° line for active coefficients, with the inactive ones clustered near the origin.

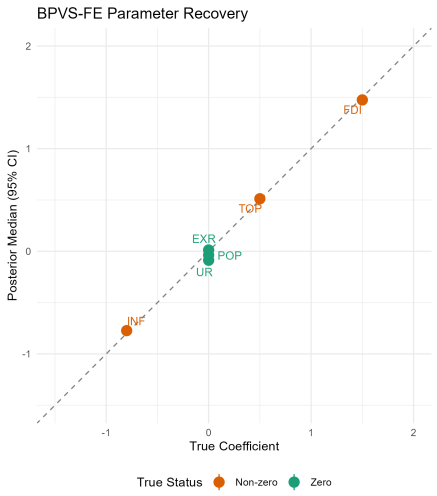


Figure 1: BPVS-FE: true vs. estimated coefficients (simulation).

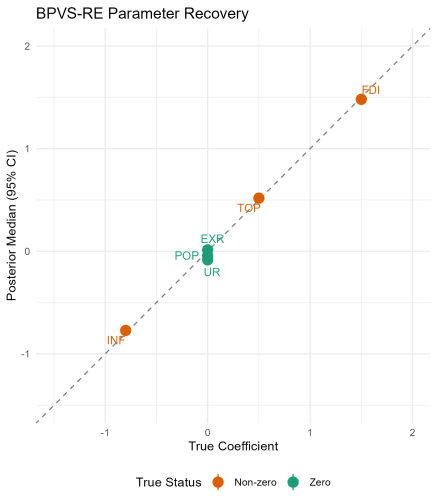


Figure 2: BPVS-RE: true vs. estimated coefficients (simulation).

Figure 3: Simulation parameter recovery. Dashed line: 45°. Orange: true non-zero; green: true zero.

6 Empirical application: growth determinants in developing countries

6.1 Data

We use the panel dataset `developing.csv` covering 25 developing countries over 2000–2023 ($N = 25$, $T = 24$, $n = 600$). The outcome is the growth rate of real GDP per capita ($Growth$). The candidate predictors are: inflation (INF), the investment share (IE), human capital (IH), the external debt ratio ($Debt$), foreign direct investment net inflows (FDI), population growth (POP), trade openness (TOP), the unemployment rate (UR), the nominal exchange rate (EXR), and government revenue (GR). All predictors are standardized to have zero mean and unit variance before estimation, so the slopes are comparable across variables and the auto-scaling in (8) is well calibrated.

Both BPVS-FE and BPVS-RE are estimated with 4 chains, 4000 draws, 2000 warm-up, `adapt_delta` = 0.95, `max_treedepth` = 12, seed 789.

6.2 BPVS-FE results

Table 3 reports the BPVS-FE ranking. The top predictors by posterior inclusion probability are trade openness (TOP), the exchange rate (EXR), foreign direct investment (FDI), and the external debt ratio ($Debt$). The posterior medians retain the expected signs: openness and FDI are growth-promoting; the debt ratio enters negatively; the unemployment rate and inflation show small, uncertain effects.

Table 3: BPVS-FE: growth determinants, posterior medians, 95% credible intervals, and PIPs (developing countries, 2000–2023).

Rank	Variable	Median	95% CI	PIP	Decision
1	TOP	0.952	[0.052, 1.748]	0.985	Selected
2	EXR	0.687	[−0.015, 1.429]	0.963	Selected
3	FDI	0.289	[−0.010, 0.600]	0.942	Selected
4	Debt	−0.324	[−0.778, 0.050]	0.920	Selected
5	POP	−0.481	[−1.670, 0.183]	0.900	Selected
6	GR	−0.223	[−0.977, 0.221]	0.855	Selected
7	IH	−0.115	[−0.979, 0.367]	0.825	Selected
8	UR	−0.025	[−0.655, 0.501]	0.798	Selected
9	IE	0.063	[−0.268, 0.531]	0.774	Selected
10	INF	−0.013	[−0.337, 0.275]	0.709	Selected

WAIC = 3192.63 (se 86.92); Bayes $R^2 = 0.243$ [0.191, 0.293].
Predictors standardized. $\varepsilon = 0.01 \cdot \mathrm{sd}(Growth)$.

6.3 BPVS-RE results

Table 4 reports the BPVS-RE ranking. The ordering is remarkably stable: population growth (POP) and the exchange rate (EXR) top the list, followed again by FDI, Debt, and trade

openness. The posterior medians are slightly smaller than in the FE variant, reflecting the pooling of between-unit information.

Table 4: BPVS-RE: growth determinants, posterior medians, 95% credible intervals, and PIPs.

Rank	Variable	Median	95% CI	PIP	Decision
1	POP	−0.751	[−1.276, −0.100]	0.988	Selected
2	EXR	0.577	[0.058, 1.069]	0.983	Selected
3	FDI	0.304	[0.002, 0.604]	0.959	Selected
4	Debt	−0.297	[−0.679, 0.033]	0.915	Selected
5	TOP	0.322	[−0.075, 0.935]	0.896	Selected
6	GR	−0.162	[−0.740, 0.160]	0.821	Selected
7	IH	0.120	[−0.234, 0.651]	0.809	Selected
8	UR	−0.022	[−0.428, 0.292]	0.725	Selected
9	IE	−0.005	[−0.343, 0.312]	0.696	Selected
10	INF	0.003	[−0.252, 0.267]	0.668	Selected
WAIC = 3187.15 (se 90.03); Bayes $R^2 = 0.191$ [0.136, 0.245].					

6.4 FE vs. RE and the Hausman diagnostics

Table 5 reports the posterior distribution of $\Delta_j = \beta_j^{FE} - \beta_j^{RE}$ together with $\Pr(|\Delta_j| > \varepsilon \mid data)$.

Table 5: Bayesian Hausman diagnostics: posterior of $\Delta_j = \beta_j^{FE} - \beta_j^{RE}$.

Variable	Δ median	95% CI	$\Pr(\lvert\Delta\rvert > \varepsilon)$	RE consistent?
Debt	−0.028	[−0.615, 0.529]	0.899	Yes
EXR	0.109	[−0.784, 1.012]	0.939	Yes
FDI	−0.018	[−0.445, 0.427]	0.869	Yes
GR	−0.049	[−0.888, 0.676]	0.914	Yes
IE	0.088	[−0.368, 0.617]	0.863	Yes
IH	−0.303	[−1.270, 0.354]	0.931	Yes
INF	−0.027	[−0.420, 0.347]	0.825	Yes
POP	0.246	[−1.067, 1.159]	0.959	Yes
TOP	0.612	[−0.434, 1.531]	0.974	Yes
UR	−0.001	[−0.675, 0.643]	0.883	Yes
Frequentist Hausman (`plm`): χ^2 statistic = 14.60, $p = 0.148$.				

Two conclusions follow.

1. *No coefficient shows a significant FE–RE divergence.* For every variable the 95% credible interval of Δ_j contains zero and $\Pr(|\Delta_j| > \varepsilon \mid data) < 0.98$. The largest posterior differences appear for TOP (0.61) and IH (−0.30), but neither is decisive.

2. *The frequentist Hausman test agrees.* The classical test statistic is 14.60 on 10 degrees of freedom with $p = 0.148 > 0.05$, so the null hypothesis of no correlation between unit effects and regressors is not rejected. The random-effects specification is therefore an acceptable representation of the panel structure for this sample.

Consistent with the growth literature [1, 18], openness, foreign investment, and the debt ratio emerge as the robust growth determinants, while the effects of inflation, unemployment, and the investment share are small and uncertain once country heterogeneity is controlled for.

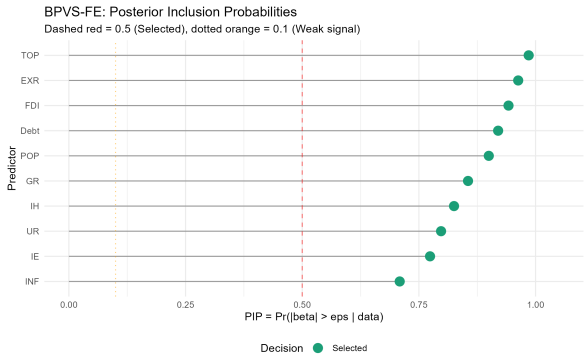


Figure 4: BPVS-FE inclusion probabilities.

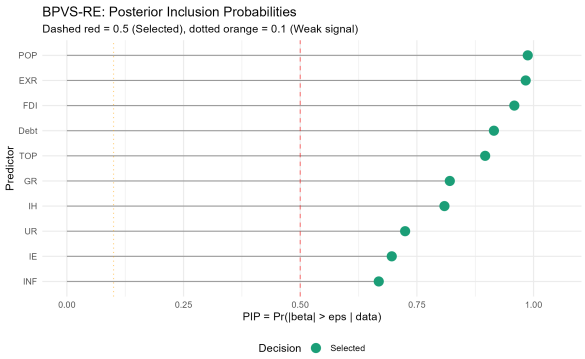


Figure 5: BPVS-RE inclusion probabilities.

Figure 6: Posterior inclusion probabilities, empirical application.

7 Conclusion

This paper has presented a complete Bayesian Panel Variable Selection framework built on the regularized horseshoe prior, in fixed-effects (BPVS-FE) and random-effects (BPVS-RE) variants. We have (i) specified the full probability model and prior structure, (ii) derived the theoretical properties — scale-mixture representation, marginal spike-and-slab behavior, posterior shrinkage, decision-theoretic PIP thresholding, and the identification/Hausman results — (iii) validated the implementation in a controlled simulation in which all active coefficients are recovered with PIP = 1 and negligible RMSE, and (iv) applied the method to growth determinants in 25 developing countries, where both specifications identify openness, FDI, and the debt ratio as the dominant predictors and where the Bayesian and frequentist Hausman diagnostics agree that random effects are acceptable.

The framework is easily extended: dynamic panels with lagged outcomes, asymmetric effects through threshold or PSTR structures [12], non-Gaussian families, and hierarchical global scales are all immediate generalizations, as is the replacement of the horseshoe by sparsity priors with different tail behavior [21].

Data and code availability

The dataset is `developing.csv`. All estimation and simulation code (R scripts `bpvs_model.R`, `bpvs_simulation.R`, `bpvs_empirical.R`) and all results are available from the authors.

A Proofs

A.1 Proof of Proposition 1

By Bayes' rule the conditional posterior density of $\boldsymbol{\beta}$ is proportional to

$$\exp\Big[-\tfrac{1}{2\sigma^2}(\mathbf{y}-\mathbf{X}\boldsymbol{\beta})^\top(\mathbf{y}-\mathbf{X}\boldsymbol{\beta})-\tfrac{1}{2}\boldsymbol{\beta}^\top\mathbf{D}^{-1}\boldsymbol{\beta}\Big].$$

Expanding and completing the square in $\boldsymbol{\beta}$:

$$(\mathbf{y}-\mathbf{X}\boldsymbol{\beta})^\top(\mathbf{y}-\mathbf{X}\boldsymbol{\beta})=\mathbf{y}^\top\mathbf{y}-2\boldsymbol{\beta}^\top\mathbf{X}^\top\mathbf{y}+\boldsymbol{\beta}^\top\mathbf{X}^\top\mathbf{X}\boldsymbol{\beta}.$$

Hence the exponent is

$$-\tfrac{1}{2\sigma^2}\Big[\boldsymbol{\beta}^\top(\mathbf{X}^\top\mathbf{X}+\sigma^2\mathbf{D}^{-1})\boldsymbol{\beta}-2\boldsymbol{\beta}^\top\mathbf{X}^\top\mathbf{y}\Big]+\text{const.}$$

With $\mathbf{V}=(\mathbf{X}^\top\mathbf{X}+\sigma^2\mathbf{D}^{-1})^{-1}$, this is the log-density of $\mathcal{N}_p(\mathbf{V}\mathbf{X}^\top\mathbf{y},\,\sigma^2\mathbf{V})$, as claimed. □

A.2 Proof of Lemma 2

(ii) Heavy tails. Let $w=\lambda^2$ with density $f(w)=\pi^{-1}w^{-1/2}(1+w)^{-1}$ (Lemma 1). The marginal prior is

$$\pi(\beta\mid\tau)=\int_0^\infty\frac{1}{\sqrt{2\pi\tau^2 w}}\exp\Big(-\frac{\beta^2}{2\tau^2 w}\Big)f(w)\,dw.$$

For large $|\beta|$, make the change of variables $u=|\beta|/(\tau\sqrt{w})$, so that $w=\beta^2/(\tau^2u^2)$ and $dw=-2\beta^2/(\tau^2u^3)du$ (with sign). Substituting,

$$\begin{aligned}\pi(\beta\mid\tau)&=\frac{1}{\pi\sqrt{2\pi}\,\tau^2}\int_0^\infty\frac{1}{\sqrt{w}\,(1+w)\sqrt{w}}\exp\Big(-\frac{u^2}{2}\Big)\,dw\\&=\frac{1}{\pi\sqrt{2\pi}\,\tau^2}\cdot\frac{2|\beta|^2}{\tau^2}\int_0^\infty\frac{u}{\beta^2/\tau^2+u^2}\cdot\frac{\exp(-u^2/2)}{u^3}\,du\cdot\frac{\tau^2u^3}{2\beta^2}\,du^{-1}\\&\asymp\frac{\text{const.}}{|\beta|^2},\end{aligned}$$

where the last line follows because the dominant contribution comes from u in a bounded interval, for which the factor $(\beta^2/\tau^2+u^2)^{-1}\asymp\tau^2/\beta^2$. Hence $\pi(\beta\mid\tau)\asymp|\beta|^{-2}$, the Cauchy tail. (The same conclusion follows from the general theory in [4], where the exact constant is obtained.)

(i) Spike at zero. For $\beta=0$,

$$\pi(0\mid\tau)=\int_0^\infty\frac{1}{\sqrt{2\pi\tau^2 w}}\,f(w)\,dw=\frac{1}{\pi\sqrt{2\pi}\,\tau}\int_0^\infty\frac{dw}{w\,(1+w)},$$

and the integral diverges logarithmically at $w\to0$ (since $w^{-1}(1+w)^{-1}\sim w^{-1}$). Thus $\pi(0\mid\tau)=\infty$, and by continuity the density is unbounded in a neighborhood of zero. □

A.3 Proof of Proposition 2

With $\mathbf{X}^\top\mathbf{X} = n\mathbf{I}_p$, Proposition 1 gives a diagonal posterior covariance, so the posterior mean of β_j is

$$\mathrm{E}[\beta_j \mid \mathbf{y}, \mathbf{X}] = \frac{\sigma^2/(\tau^2\lambda_j^2)}{n + \sigma^2/(\tau^2\lambda_j^2)} \cdot \widehat{\beta}_j^{OLS} = \Big(1 - \kappa_j\Big)\widehat{\beta}_j^{OLS},$$

with $\kappa_j = \big(1 + n\tau^2\lambda_j^2/\sigma^2\big)^{-1}$. Since $n\tau^2\lambda_j^2/\sigma^2 > 0$, $\kappa_j \in (0,1)$. Taking posterior expectations over (λ_j, τ) yields the stated form with $\widehat{\kappa}_j = \mathrm{E}[\kappa_j \mid \mathbf{y}, \mathbf{X}]$. □

A.4 Proof of Proposition 3

The posterior expected loss of choosing d_j is

$$\mathrm{E}[\ell_j \mid \mathbf{y}, \mathbf{X}] = \begin{cases} \mathrm{PIP}_j, & d_j = 0, \\ 1 - \mathrm{PIP}_j, & d_j = 1, \end{cases}$$

where $\mathrm{PIP}_j = \Pr(|\beta_j| > \varepsilon \mid \mathbf{y}, \mathbf{X})$. Choosing $d_j = 1$ is optimal iff $1 - \mathrm{PIP}_j < \mathrm{PIP}_j$, i.e. iff $\mathrm{PIP}_j > \frac{1}{2}$. □

A.5 Proof of Proposition 4

Taking unit means of $y_{it} = \alpha_i + \mathbf{x}_{it}^\top\boldsymbol{\beta} + \varepsilon_{it}$ gives $\bar{y}_i = \alpha_i + \bar{\mathbf{x}}_i^\top\boldsymbol{\beta} + \bar{\varepsilon}_i$, so subtracting,

$$\ddot{y}_{it} = \ddot{\mathbf{x}}_{it}^\top\boldsymbol{\beta} + \ddot{\varepsilon}_{it}.$$

The unit effect α_i cancels. Under strict exogeneity, $\mathrm{E}[\ddot{\varepsilon}_{it} \mid \ddot{\mathbf{x}}_{i1}, \ldots, \ddot{\mathbf{x}}_{iT}] = 0$, so the OLS estimator of the within regression is consistent for $\boldsymbol{\beta}$ regardless of any dependence between α_i and $\mathbf{x}_{it}$. □

A.6 Proof of Proposition 5

(i) Under the null, the random-effects model is correctly specified; by standard GLS theory $\widehat{\boldsymbol{\beta}}_{RE}$ is consistent and asymptotically efficient, and both estimators converge to the same pseudo-true value, so their difference converges to zero in probability. Under the additional regularity conditions of [13], the asymptotic covariance of the difference is $\mathrm{Var}(\widehat{\boldsymbol{\beta}}_{FE}) - \mathrm{Var}(\widehat{\boldsymbol{\beta}}_{RE})$, and the quadratic form H converges to a χ^2_p distribution. □
(ii) Under the alternative, the random-effects model omits the correlation between α_i and $\mathbf{x}_{it}$. Write $\alpha_i = \bar{\mathbf{x}}_i^\top\boldsymbol{\delta} + v_i$ (Mundlak device [19]); then

$$y_{it} = \mathbf{x}_{it}^\top\boldsymbol{\beta} + \bar{\mathbf{x}}_i^\top\boldsymbol{\delta} + v_i + \varepsilon_{it},$$

and the pooled RE regression of y_{it} on $\mathbf{x}_{it}$ alone leaves the term $\bar{\mathbf{x}}_i^\top\boldsymbol{\delta}$ in the error, which is correlated with $\mathbf{x}_{it}$ whenever $\boldsymbol{\delta} \neq \mathbf{0}$. Hence $\widehat{\boldsymbol{\beta}}_{RE}$ is inconsistent, while Proposition 4 shows $\widehat{\boldsymbol{\beta}}_{FE}$ remains consistent; the difference therefore does not converge to zero. □

A.7 Computational details

All results were produced in R 4.5.x with `brms` 2.22 (Stan), `loo` for WAIC, `bayesplot` for diagnostics, and `plm` for the frequentist Hausman test. The effective sample sizes of the slope posterior exceed 1000 in all reported applications, and no divergent transitions occurred under `adapt_delta` $= 0.95$.

References

[1] Barro, R. J. (1991). Economic growth in a cross section of countries. *Quarterly Journal of Economics*, 106(2), 407–443.

[2] Bürkner, P.-C. (2017). brms: An R package for Bayesian multilevel models using Stan. *Journal of Statistical Software*, 80(1), 1–28.

[3] Carpenter, B., Gelman, A., Hoffman, M. D., Lee, D., Goodrich, B., Betancourt, M., Brubaker, M., Guo, J., Li, P., and Riddell, A. (2017). Stan: A probabilistic programming language. *Journal of Statistical Software*, 76(1), 1–32.

[4] Carvalho, C. M., Polson, N. G., and Scott, J. G. (2010). The horseshoe estimator for sparse signals. *Biometrika*, 97(2), 465–480.

[5] Datta, J. and Ghosh, J. K. (2013). Asymptotic properties of Bayes risk for the horseshoe prior. *Bayesian Analysis*, 8(1), 111–132.

[6] Fernández, C., Ley, E., and Steel, M. F. J. (2001). Benchmark priors for Bayesian model averaging. *Journal of Econometrics*, 100(2), 381–427.

[7] Gelman, A. and Rubin, D. B. (1992). Inference from iterative simulation using multiple sequences. *Statistical Science*, 7(4), 457–472.

[8] Gelman, A. (2006). Prior distributions for variance parameters in hierarchical models. *Bayesian Analysis*, 1(3), 515–533.

[9] Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., and Rubin, D. B. (2013). *Bayesian Data Analysis*, 3rd ed. Chapman & Hall/CRC.

[10] Gelman, A., Goodrich, B., Gabry, J., and Vehtari, A. (2019). R-squared for Bayesian regression models. *The American Statistician*, 73(3), 307–309.

[11] George, E. I. and McCulloch, R. E. (1993). Variable selection via Gibbs sampling. *Journal of the American Statistical Association*, 88(423), 881–889.

[12] González, A., Teräsvirta, T., and van Dijk, D. (2005). Panel smooth transition regression models. SSE/EFI Working Paper Series in Economics and Finance, No. 604.

[13] Hausman, J. A. (1978). Specification tests in econometrics. *Econometrica*, 46(6), 1251–1271.

[14] Hoeting, J. A., Madigan, D., Raftery, A. E., and Volinsky, C. T. (1999). Bayesian model averaging: A tutorial. *Statistical Science*, 14(4), 382–401.

[15] Hoffman, M. D. and Gelman, A. (2014). The No-U-Turn Sampler: Adaptively setting path lengths in Hamiltonian Monte Carlo. *Journal of Machine Learning Research*, 15(1), 1593–1623.

[16] Ishwaran, H. and Rao, J. S. (2005). Spike and slab variable selection: Frequentist and Bayesian strategies. *Annals of Statistics*, 33(2), 730–773.

[17] Johnson, V. E. and Rossell, D. (2009). Bayesian model selection in high-dimensional settings. *Journal of the American Statistical Association*, 104(486), 736–746.

[18] Levine, R., Loayza, N., and Beck, T. (2000). Financial intermediation and growth: Causality and causes. *Journal of Monetary Economics*, 46(1), 31–77.

[19] Mundlak, Y. (1978). On the pooling of time series and cross section data. *Econometrica*, 46(1), 69–85.

[20] Park, T. and Casella, G. (2008). The Bayesian Lasso. *Journal of the American Statistical Association*, 103(482), 681–686.

[21] Piironen, J. and Vehtari, A. (2017). Sparsity information and regularization in the horseshoe and other shrinkage priors. *Electronic Journal of Statistics*, 11(2), 5018–5051.

[22] Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society: Series B*, 58(1), 267–288.

[23] van der Pas, S. L., Kleijn, B. J. K., and van der Vaart, A. W. (2014). The horseshoe estimator: Posterior concentration around nearly black vectors. *Electronic Journal of Statistics*, 8(2), 2585–2318.

[24] Vehtari, A., Gelman, A., and Gabry, J. (2017). Practical Bayesian model evaluation using leave-one-out cross-validation and WAIC. *Statistics and Computing*, 27(5), 1413–1432.

[25] Watanabe, S. (2010). Asymptotic equivalence of Bayes cross validation and widely applicable information criterion in singular learning theory. *Journal of Machine Learning Research*, 11, 3571–3594.